

WALIYA SHAH

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PROFILE

- Data professional with 5+ years of experience delivering scalable analytics, enterprise ETL pipelines, and predictive models across energy, retail, and healthcare industries.
- Advanced expertise in Python, SQL, AWS, and BI systems, automating 50+ workflows and supporting datasets exceeding 2M records/month.
- Proven success building centralized data platforms, orchestrating end-to-end pipelines (Airflow, Glue, Docker), and enabling insights for 50+ operational stakeholders.
- Strong background in KPI engineering, dashboard design, and data validation automation, elevating reporting accuracy up to 99%.
- Recognized for transforming manual business processes into automated, reliable, and data-driven solutions that improve performance and decision-making.

TECHNICAL SKILLS

Programming & Data Science: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit), SQL (T-SQL, PL/SQL), VBA, Flask, REST APIs, Git, Jupyter, Google Colab

Data Engineering & Pipelines: ETL Development, Data Pipelines, Airflow, AWS Glue, DBT, Docker, CI/CD, Data Validation, Great Expectations, Pipeline Monitoring, Web Scraping, API Integration

Databases & Warehousing: SQL Server, PostgreSQL, MySQL, Redshift, Data Modeling, Query Optimization, Indexing, Stored Procedures, Data Architecture, Performance Tuning

Business Intelligence & Reporting: Power BI (DAX, Power Query), Tableau, Looker, Excel Power Tools, KPI Dashboarding, Reporting Automation, Data Visualization, Executive Reporting

Cloud & Infrastructure: AWS (S3, Redshift, Glue, EC2, Lambda, IAM), Docker, Terraform, Version Control, Containerized Workflows

Analytics & Statistical Modeling: EDA, Data Cleaning, Hypothesis Testing, Regression, Classification, Feature Engineering, Predictive Modeling, Model Evaluation, A/B Testing, Data Governance

PROFESSIONAL EXPERIENCE

Research Assistant - Python Automation & SSI Developer Aug 2025 - Present
Western New England University Springfield, MA

- Engineered automated credential-management workflows using Hyperledger Aries and Python, enabling secure issuance, verification, and revocation for 1,000+ academic and healthcare credentials.
- Containerized multi-agent Aries workflows in Docker and orchestrated REST API interactions, improving processing reliability by 40% and reducing manual steps by 60%.
- Integrated decentralized identity protocols into internal systems, strengthening authentication accuracy and reducing credential validation time by 50%.

Data Analyst Intern - Energy Supply Department Jan 2025 - May 2025
Westfield Gas & Electric Westfield, MA

- Designed SQL Server-based data models and ETL workflows to unify legacy spreadsheets into a centralized BI platform supporting 5+ departments and 40+ active users.
- Automated monthly energy-supply KPI pipelines using Python and SQL, eliminating 90% manual workload (30+ labor hours/month) and improving reporting accuracy to 99%.
- Built internal analytics dashboards and data summaries to support operational decision-making, enhancing visibility into energy usage trends, department-level performance, and compliance-related metrics.
- Integrated Flask APIs and automated email alerts for workflow approvals, reducing reporting delays by 70% and strengthening organizational accountability.
- Partnered with the KRA Subcommittee to align KPIs, ensuring consistent data governance and compliance with municipal and regulatory standards.

Research Assistant - Data Analyst May 2024 - Aug 2024
Western New England University Springfield, MA

- Analyzed 100K+ chatbot interaction logs to optimize intent recognition and response accuracy, improving model performance by approximately 70%.
- Developed scalable ETL pipelines that processed 20K+ FDA medical records monthly, boosting data accessibility for research teams by 85%.
- Built secure REST APIs using Swagger and Python for structured data access across research workflows, reducing retrieval time by 60%.
- Created trend analytics and statistical visualizations (Matplotlib, Seaborn) to detect adverse event patterns, enhancing research insight quality by 80%.
- Optimized PostgreSQL database structures and queries, improving read/write performance by 90%.

Research Assistant - Python Programmer Mar 2024 - Jul 2024
Western New England University Springfield, MA

- Deployed decentralized identity verification workflows using Hyperledger Aries and ACA-Py, supporting secure agent communication across multiple research applications.
- Automated credential issuance and proof validation processes using Python modules and Dockerized micro-agents, decreasing validation latency by 40%.
- Developed Swagger/OpenAPI integrations that standardized API documentation and improved team onboarding efficiency by 35%.

Senior Data Analyst Jan 2020 - May 2023
SnappRetail Pakistan

- Built 20+ executive Tableau dashboards covering sales, inventory, pricing, and demand-forecast KPIs, used daily by 50+ business, merchandising, and retail stakeholders.
- Automated 30+ recurring reports using Python and Excel macros, saving 50 hours/month and improving data accuracy and operational transparency.
- Designed and maintained AWS-based ETL pipelines (S3, Redshift, Glue) to ingest, clean, and transform 2M+ monthly POS transactions, reducing data discrepancies by 80% and improving forecasting reliability.
- Developed competitor-price web scrapers monitoring 500+ SKUs weekly, enabling strategic pricing decisions and improving margin stability.
- Implemented data-validation rules, error-logging checks, and automated alerts for pipeline failures, improving data quality and reducing manual intervention by 35%.
- Collaborated with merchandising, supply chain, and finance teams to refine KPIs, define reporting requirements, and support business reviews and strategic planning.
- Led the migration of legacy Excel-based processes to SQL and tableau-driven workflows, reducing reporting turnaround time from hours to minutes.

Data Analyst Intern Mar 2019 - Dec 2019
K-Labs Pakistan

- Developed KPI and trend reports using Excel Power Query, Power Pivot, and PivotTables to support client analytics needs across marketing and operations.
- Automated 10+ recurring workflows using advanced formulas and macros (VLOOKUP, INDEX-MATCH), reducing reporting turnaround time by 60%.
- Conducted data cleaning and transformation tasks supporting dashboard development and metric accuracy improvements.

CERTIFICATIONS & PROJECTS

Certifications: Certified Data Analyst (FAST-NU) · Google Data Analytics (Coursera) · Web Dev with Flask · BI Dashboard with BigQuery (Google Cloud)

Projects: Streamlit App (Python) · NLP Text Analysis (Scikit-learn) · HR Dashboard

EDUCATION

Master of Science in Computer Science May 2025
Western New England University Springfield, MA